

PROG7314 - Programming 3D

Part 1 - Research, Planning and Design

ST10084625 - Caleb Keanu Naidoo

ST10441678 - Taytem Pillay

ST10460086 - Reece Mikhael Moodley



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SECTION 1: RESEARCH REPORT

1. Introductions

The contemporary higher education landscape presents a unique set of operational and social dynamics that generic commercial applications fail to navigate effectively. Tertiary students operate within highly synchronized academic schedules, localized geographic perimeters, and specific budget constraints. They require specialized platforms for peer-to-peer commerce, verified academic resource sharing, and structured skill collaboration. Traditional classified platforms and generalized social media marketplaces prioritize broad consumer reach over localized trust, leading to systemic vulnerabilities including unverified profile proliferation, predatory pricing, scam operations, and severe search noise when attempting to locate course-specific materials.

To address these architectural and functional gaps, this research report conducts an empirical, comparative evaluation of three representative marketplace platforms: Gumtree, Facebook Marketplace, and StudentVIP. By analyzing their architectural paradigms, state persistence engines, user interface design patterns, and trust verification mechanisms, this study extracts key technical insights. The primary objective is to establish an evidence-based foundation for engineering Matrix Market a hyper-localized, native Android application tailored specifically to the needs of university communities.

2. App 1 Research: Gumtree

2.1 Overview & Target Audience

Gumtree represents one of the earliest and most widely adopted online classifieds platforms. Operating across broad nationwide markets, its primary target audience encompasses a general demographic spanning casual individual buyers, high-volume second-hand sellers, and commercial vendors (Africa, 2026). The platform prioritizes high item volume and geographic radius discovery over identity verification or community-specific categorization.

2.2 Functional Strengths & Weaknesses

- Strengths:
 - Comprehensive macro-level categorization tree that accommodates virtually all physical goods, motor vehicles, property rentals, and general service types.

- Flexible location-based search filtering operating on configurable municipal kilometre radii to locate items within immediate geographical vicinity.
- Multiple communication channels allowing users to initiate contact via direct telephone calls, external SMS, or internal asynchronous messaging.
- High organic user traffic volume and massive catalogue saturation ensuring maximum initial exposure for common non-specialized consumer items.
- Saved search alert configurations that dispatch automated push notifications whenever new inventory matching specific query keywords enters the database.
- Weaknesses:
 - Complete absence of institutional identity verification mechanisms, rendering the platform highly susceptible to fraudulent accounts, phishers, and automated spam bots.
 - Highly intrusive third-party advertisement integration that congests visual layouts, increases initial screen render latency, and disrupts core user workflows.
 - Total inability to organize listings by academic taxonomy, such as university module codes, textbook ISBN identifiers, or campus-specific drop-off zones.
 - Absence of built-in escrow or secure payment processing engines, forcing users into unverified cash transactions or unsafe off-platform bank transfers.
 - Outdated, unstructured search indexing that produces excessive search noise when queries involve specific academic editions or institutional course material.

2.3 Technical Architecture & State Management

Gumtree's legacy architecture relies heavily on an imperative Model-View-Presenter (MVP) pattern that has incrementally migrated toward Model-View-ViewModel (MVVM). Interface layouts are structured using traditional Android XML files with View Binding wrappers (Saraiva, 2023). This traditional view hierarchy introduces significant layout depth, leading to increased UI thread overhead during complex list rendering and smooth touch fling gestures (Developers, 2026).

State management and persistent data caching rely on reactive streams via RxJava coupled with a localized SQLite relational database. While functional under stable conditions, the synchronization pipeline between network responses and the local

cache demonstrates state inconsistency during rapid network drops, occasionally rendering stale or duplicate listing data to the user layout.

3. App 2 Research: Facebook Marketplace

3.1 Overview & Target Audience

Facebook Marketplace operates as an embedded social commerce hub within the larger Meta ecosystem. By leveraging pre-existing social graph connections, it targets active social network users seeking friction-free, localized consumer-to-consumer exchanges driven by real-time algorithmic recommendations and integrated instant messaging.

3.2 Functional Strengths & Weaknesses

- Strengths:
 - Direct linkage to active Facebook profiles, offering implicit social proof through mutual friend displays, profile account age, and publicly visible activity metrics.
 - Instantaneous, low-latency communication powered by the Meta Messenger infrastructure, supporting rich media sharing, live location sharing, and active status indicators.
 - Sophisticated machine learning discovery engines that dynamically tailor the user feed based on real-time telemetry, search history, and implicit interest mapping.
 - Seamless integration with localized social groups, allowing targeted inventory broadcasting directly into specialized buy-and-sell student communities.
 - Frictionless listing creation workflow featuring automated image object recognition and AI-driven category auto-suggestions to accelerate publishing time.
- Weaknesses:
 - High prevalence of bad-actor accounts, burner profiles, and sophisticated marketplace scams that easily bypass automated moderation filters.
 - Unstructured item categorization that forces students to execute generic keyword searches rather than filtered queries based on institutional unit codes or academic editions.

- Mandatory dependency on a full social media account, entirely excluding privacy-conscious students or those without an active social media footprint.
- Invasive background tracking and data harvesting routines that raise significant privacy concerns regarding user location telemetry and browsing habits.
- Frequent buyer ghosting and low transaction commitment due to overly frictionless quick-reply buttons that initiate low-intent automated inquiries.

3.3 Technical Architecture & State Management

Facebook Marketplace is constructed using React Native, a cross-platform framework that executes JavaScript business logic on a background thread while communicating asynchronously over a bridge to native renderers. The UI framework utilizes component-based declarative view structures. While enabling rapid feature deployment across platforms, the bridge architecture can introduce micro-stutters and frame drops during intense layout calculations, heavy image scrolling, or rapid user gesture inputs.

Data persistence and network state management are driven by GraphQL client architecture (Relay) and aggressive local key-value caching. The system heavily leverages optimistic UI updating rendering predicted state changes immediately on screen before receiving server confirmation. This creates a highly responsive feel, though it occasionally requires rollback handling when network calls fail asynchronously.

4. App 3 Research: StudentVIP

4.1 Overview & Target Audience

StudentVIP is a dedicated, domain-specific Australian tertiary platform designed specifically for university students. Its target demographic is strictly limited to higher education matriculants seeking subject-matched second-hand textbooks, accredited peer tutoring services, campus navigational maps, and course-specific advice (Australia, 2026).

4.2 Functional Strengths & Weaknesses

- Strengths:
 - Granular academic taxonomy organized precisely by institutional campus, degree program, and official subject unit code.
 - Dedicated ISBN textbook validation engine that auto-populates book metadata, edition numbers, and author details to guarantee exact item matches.
 - Specialized community support modules, including verified peer tutor directories backed by historical student reviews and grade validation.
 - Institution-filtered campus map integration assisting new tertiary students in locating specific academic halls, lecture theatres, and campus amenities.
 - Targeted audience isolation ensuring listings remain relevant strictly to university life without commercial noise or irrelevant consumer goods.
- Weaknesses:
 - Visually dated, web-centric user interface marked by high density, small tap targets, and non-intuitive touch navigation flows.
 - Total absence of real-time, in-app chat infrastructure; transactions rely on manual, asynchronous email exchanges or external phone messaging.
 - Poor mobile performance and frame lag caused by web-view wrappers, lacking fluid 60fps animations and native gesture handling.

- Complete lack of integrated push notification pipelines, resulting in delayed communication and missed transaction opportunities between buyers and sellers.
- Inflexible, static search indexes that fail to support dynamic, multi-attribute filtering across simultaneous academic parameters like module code, condition, and price range (Neil, 2014).

4.3 Technical Architecture & State Management

StudentVIP employs a hybrid architecture, embedding web content inside native platform containers (Cordova/PhoneGap layout model). The user interface is built on responsive HTML5, CSS3, and JavaScript frameworks. Because layout elements are parsed through a browser engine wrapper rather than compiled directly to native OS widgets, the user experience suffers from latency, sluggish touch response times, and inconsistent animation rendering.

State management is entirely web-dependent, relying on standard HTTP polling, cookie sessions, and local browser storage. The application exhibits virtually zero offline capability any interruption in cellular connectivity completely halts navigation, rendering cached listings unavailable until active network connectivity is restored.

5. Platform Comparison Matrix & Comprehensive Comparative Analysis

Below is the structured comparative evaluation mapping out the core architectural, functional, and user experience differences across all three platforms alongside the proposed engineering specifications for Matrix Market:

Evaluation Dimension	App 1: Gumtree	App 2: Facebook Marketplace	App 3: StudentVIP	Proposed App: Matrix Market
Primary Target Audience	General public, commercial vendors, local C2C buyers/sellers	Active social network users, local C2C buyers	University students, academic peer tutors	Tertiary students, campus peer networks
Identity Verification Architecture	Unverified (Open email/phone registration; highly vulnerable to fraudulent bot profiles and scam operations)	Social Graph Link (Implicit social proof via Facebook profile history; vulnerable to burner accounts and privacy intrusion)	Domain Email (.edu address validation; isolates access to verified students but lacks real-time biometric or student ID validation)	Multi-Factor Onboarding: Institutional SSO + .edu email domain confirmation + Camera Optical Biometric / Student ID Card parsing
UI Framework & Performance	Legacy Native (Android XML View Binding; deep layout hierarchies causing visual thread measurement overhead and frame drops)	Cross-Platform (React Native Bridge; asynchronous JavaScript bridge overhead causing micro-stutters during heavy scrolling)	Hybrid Web-View (Cordova / HTML5 Wrapper; web engine parsing causing sluggish touch response and non-native rendering)	Modern Declarative Native (Kotlin + Jetpack Compose; direct compilation to GPU commands delivering steady 60fps performance)
Catalog Taxonomy &	Macro-level commercial	Algorithmic feed + general	Academic taxonomy	Academic Taxonomy +

Search Mechanics	categories (General search engine producing severe search noise for academic materials)	categories (Serendipitous discovery engine; poor targeting for specific textbook editions or unit codes)	(Organized by Campus, Unit Code, and ISBN; structured cataloguing but static multi-attribute filtering)	Camera ISBN Barcode Scanner (ML Kit) + Multi-Attribute Filtering (Module Code, Condition, Campus, Skill-Link)
Data Persistence & Offline Resilience	SQLite + RxJava (Partial local cache; synchronization latency during network drops leads to stale data renders)	GraphQL Client Store / Relay (Optimistic state updating; highly responsive but requires state rollbacks on async network failure)	Web LocalStorage (Zero offline capabilities; live HTTP polling completely halts navigation during loss of connectivity)	Local-First Architecture: Room Database + Kotlin Coroutines & Flows (Single source of truth providing total offline browsing and background sync)
Communication Infrastructure	Async internal chat, exposed SMS/Phone (No transaction tracking, seller protection, or verified receipt logging)	Real-time Meta Messenger integration (Low-latency messaging with rich media; high rate of buyer ghosting via quick-reply buttons)	External email & manual phone contact (No built-in real-time messaging engine; severe communication lag between students)	Real-Time WebSocket & Firebase Chat Pipeline (In-app negotiation without disclosing personal social profiles or contact numbers)
Trust & Reputational Systems	Basic 5-star seller rating (Unverified public reviews prone to manipulation or retaliatory ratings)	Public profile age & rating feedback (Dependent on broader social network activity rather than	Subject grade verification & tutor reviews (Targeted academic proof for peer tutors)	Gamified Student Score System (Earned reputational points, verified trade badges, academic trust tiers, and response metrics)

		transaction compliance)		
Monetization & Visual Layout	High Ad Density (Intrusive third-party banner ads that congest layouts and increase initial screen render latency)	Moderate Ad Density (Sponsored algorithmic feed ads seamlessly embedded into product discovery streams)	Low Ad Density (Targeted educational sponsor links and academic promotional banners)	Zero Intrusive Ads (Clean visual layout prioritizing student items; optional premium student perks)

6. Feature Adoption & Technical Strategy for Matrix Market

To resolve the engineering and UX flaws identified across Gumtree, Facebook Marketplace, and StudentVIP, Matrix Market will execute a targeted, native-first development strategy utilizing modern Android standards:

1. Mandatory Institutional Trust Pipeline:
 - Derived from: StudentVIP's academic scope.
 - Engineering Strategy: Implement multi-tiered onboarding requiring institutional student email (.edu/academic domain) verification combined with facial biometrics or student ID card optical parsing. This entirely eradicates the unverified bot activity present on Gumtree.
2. Declarative Native Engineering (Jetpack Compose + MVVM):
 - Addressing: Gumtree's heavy XML view hierarchies and StudentVIP's sluggish web-views.
 - Engineering Strategy: Build the application exclusively in Kotlin using Jetpack Compose following MVVM + Clean Architecture. Jetpack Compose compiles directly to native GPU commands, eliminating JavaScript bridges (Facebook Marketplace) or web-view overhead (StudentVIP) to guarantee fluid 60fps performance (Saraiva, 2023) (Developers, 2026).
3. Offline-First Reactive Data Architecture (Room + Kotlin Flows):
 - Addressing: StudentVIP's total lack of offline capability and Gumtree's state synchronization lag.
 - Engineering Strategy: Deploy a single source of truth data flow utilizing Room Local Database integrated with Kotlin Coroutines and Flows. All

network fetches immediately write to the local Room cache before emitting state to the UI, enabling instant, offline-capable browsing with background cloud synchronization.

4. Structured Campus Taxonomy & ISBN Barcode Engine:

- Derived from: StudentVIP's course-centric cataloging.
- Engineering Strategy: Integrate a camera-driven barcode scanner engine (ML Kit) that reads physical textbook ISBNs and maps them instantly to ASP.NET Core backend databases pre-populated with university-specific module codes (Neil, 2014).

5. Student Score & Peer Engagement Mechanics:

- Derived from: Facebook Marketplace's social proof mechanisms.
- Engineering Strategy: Introduce a custom gamified trust system for example Student Score. Users earn reputational score points by completing verified transactions, verifying academic status, and maintaining high response rates (Xiong, 2004).

7. Conclusion

The comparative research demonstrates a clear market void: generic platforms offer high technical performance and real-time features but lack student safety and academic organization, whereas student-focused platforms provide domain-specific categorization but suffer from outdated hybrid web architectures and non-existent offline persistence. Matrix Market bridges this technical and operational gap by merging modern native Android architecture for example Kotlin, Jetpack Compose, Room, MVVM, with institutional identity verification, specialized course taxonomies, and interactive student engagement features.

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